

Scare You

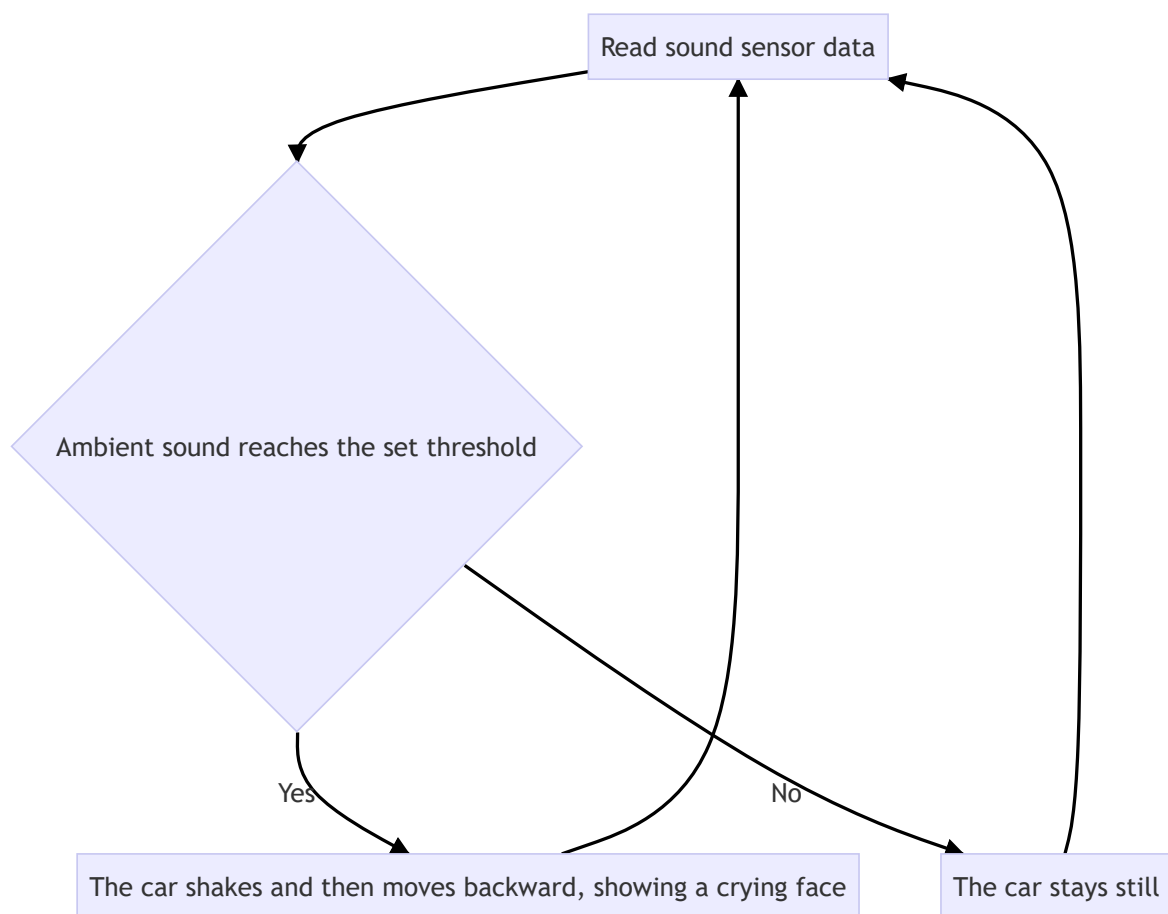
Scare You

1. Experiment Purpose
2. Implementation Principle
3. Experiment Steps
4. Experiment Phenomenon

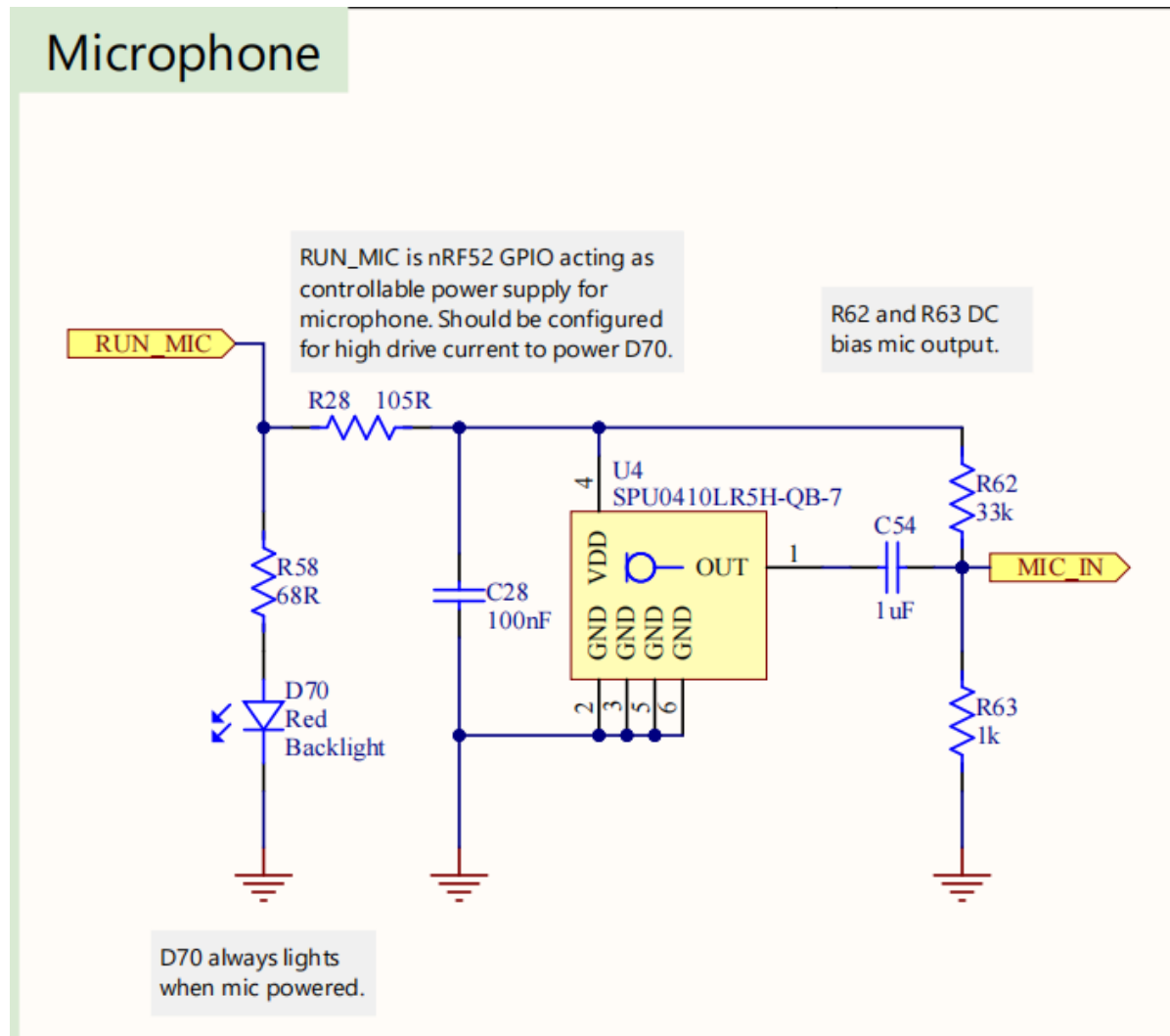
1. Experiment Purpose

In this section, we will learn to use the microphone sensor to detect sound and achieve the "scare you" effect. When a loud sound is detected in the environment, the car will show a "scared" reaction: displaying a sad expression, shaking its body, moving backward, and stopping.

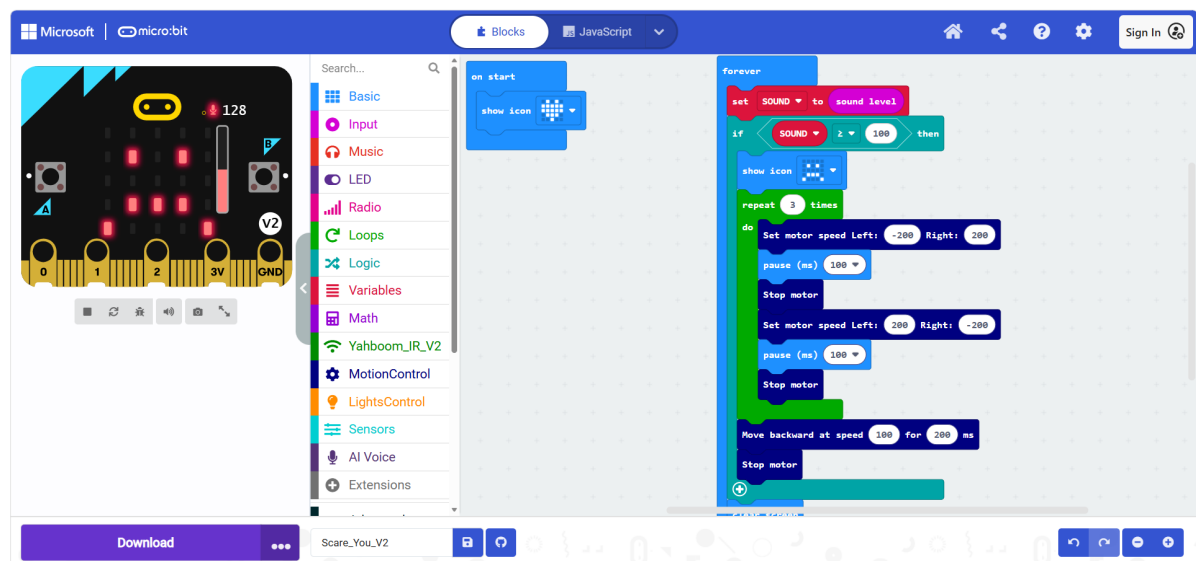
2. Implementation Principle



A microphone sensor is provided near the Logo of the Micro:bit V2. When the program drives and reads it, the red LED lights up; when it is not read, the red LED turns off and the microphone is shut down, which can reduce the power consumption of the Micro:bit at the hardware level.



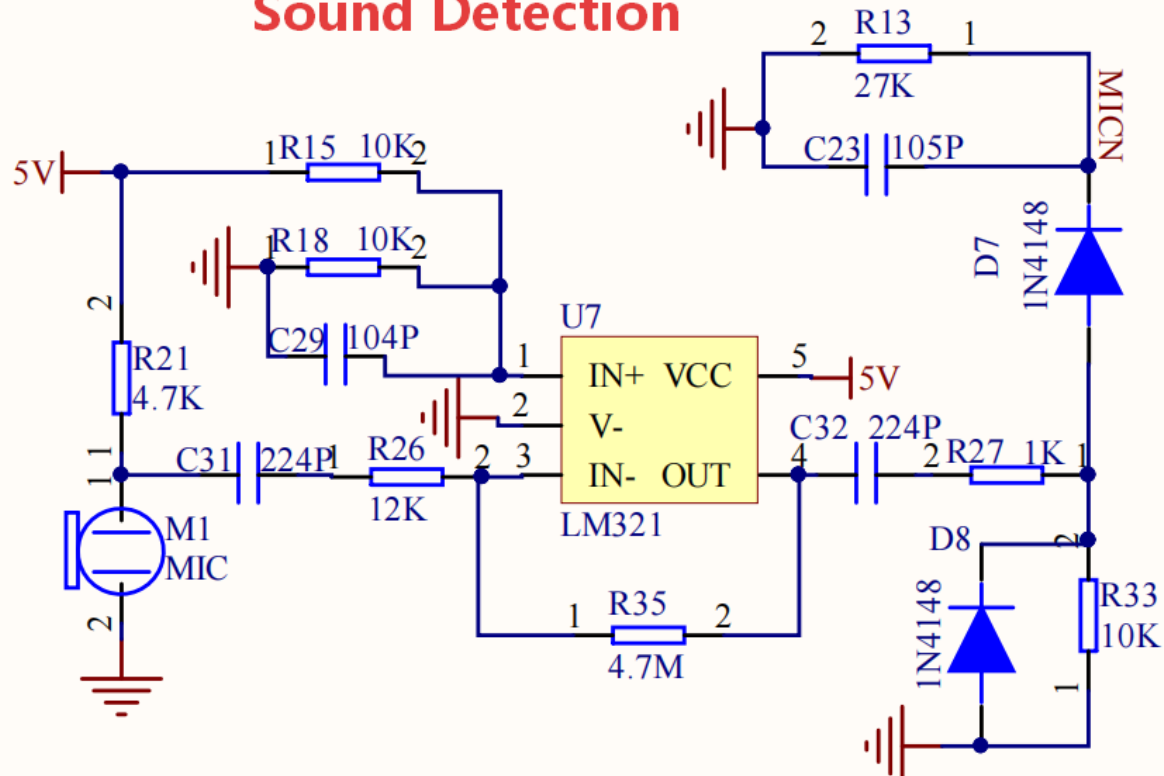
V2 version program



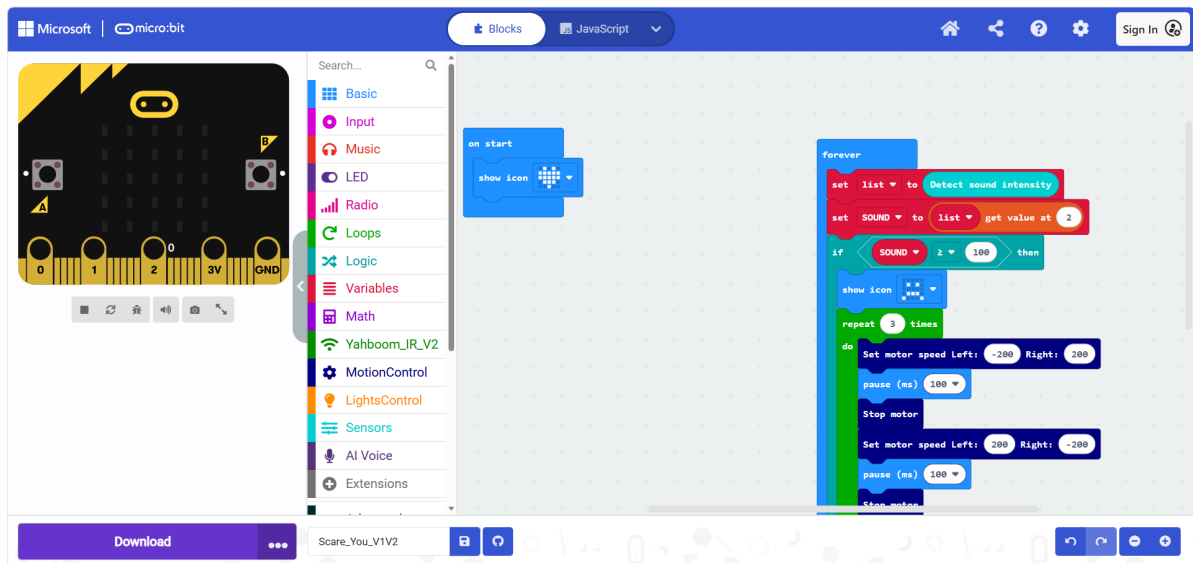
V1V2 version (expansion board MIC sensor)

The on-board sound sensor consists of an electret microphone (commonly called a mic head) plus signal amplification and clamping protection, forming a complete sound detection circuit.

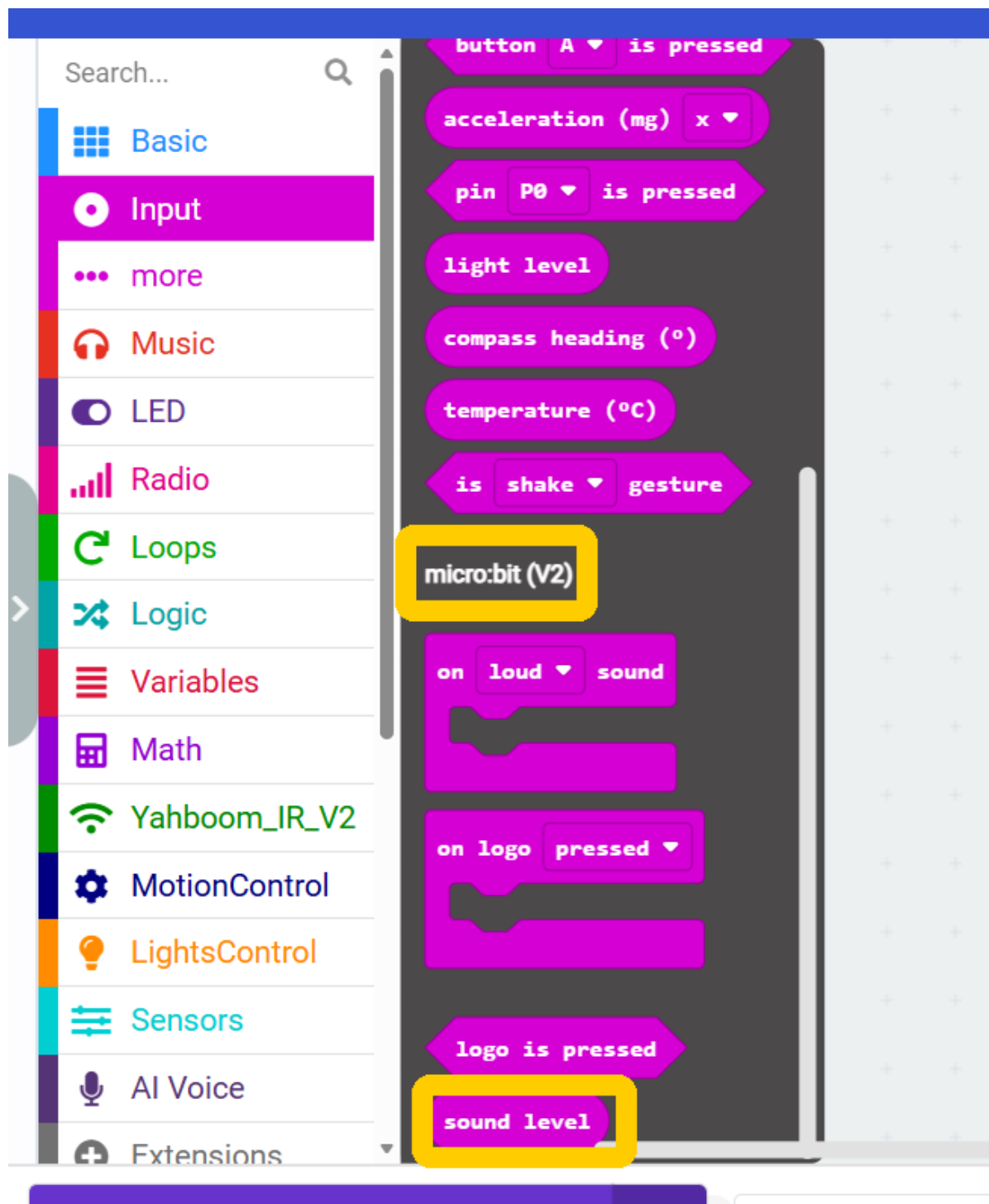
Sound Detection



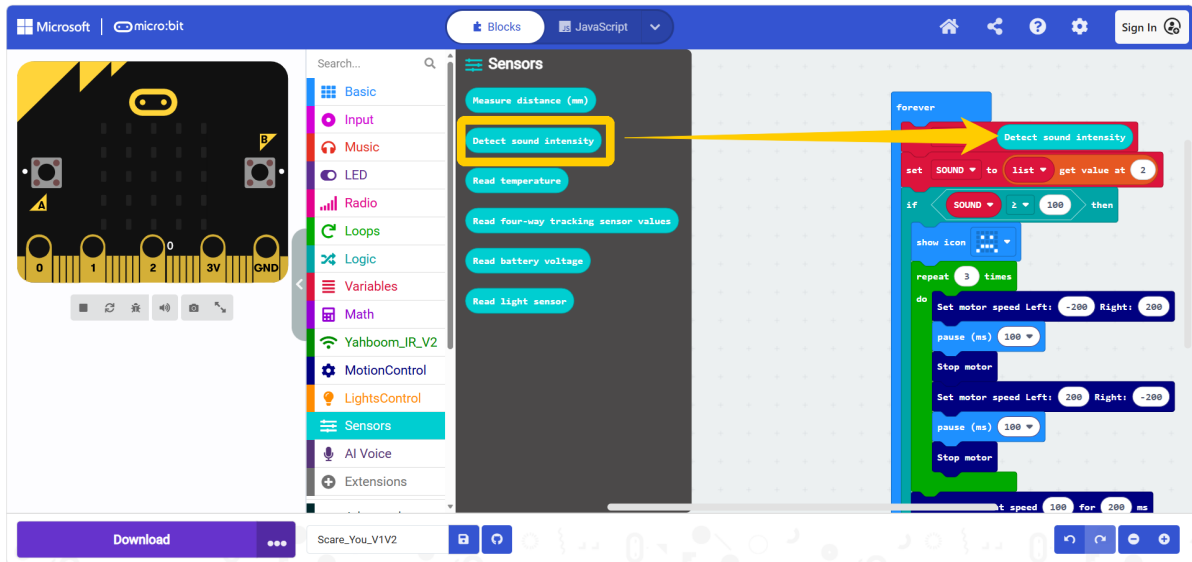
V1V2 version program



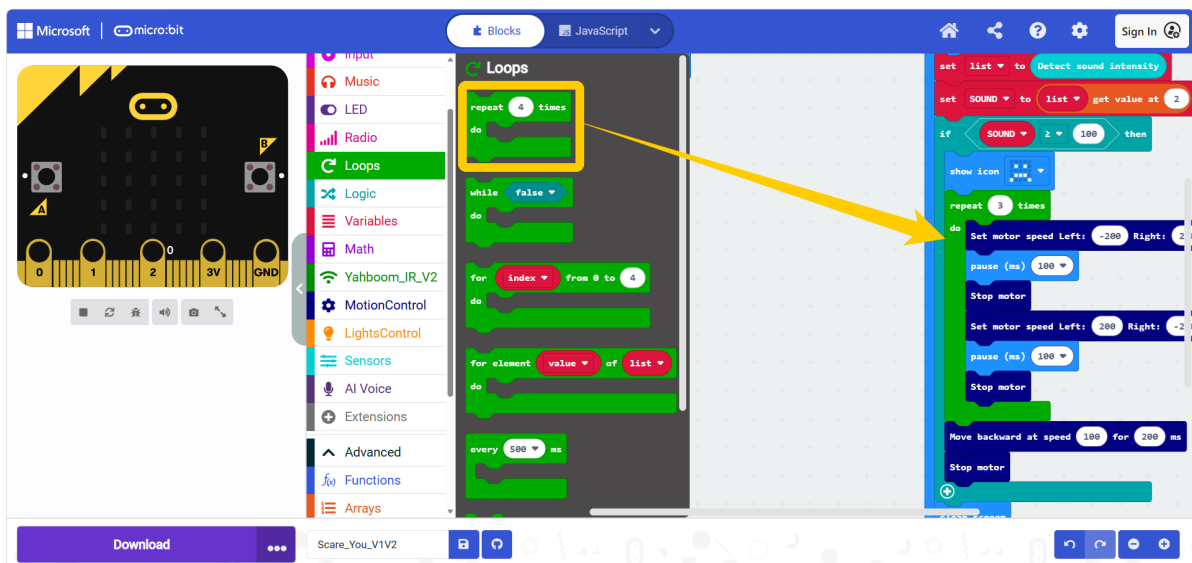
The V2 version reads the sound loudness in the `Microbit v2` subcategory of the `Input` category.



The V1V2 version reads the sound loudness in the `Sensors` category.



The repeat (n) times block is located in the **Logic** category.



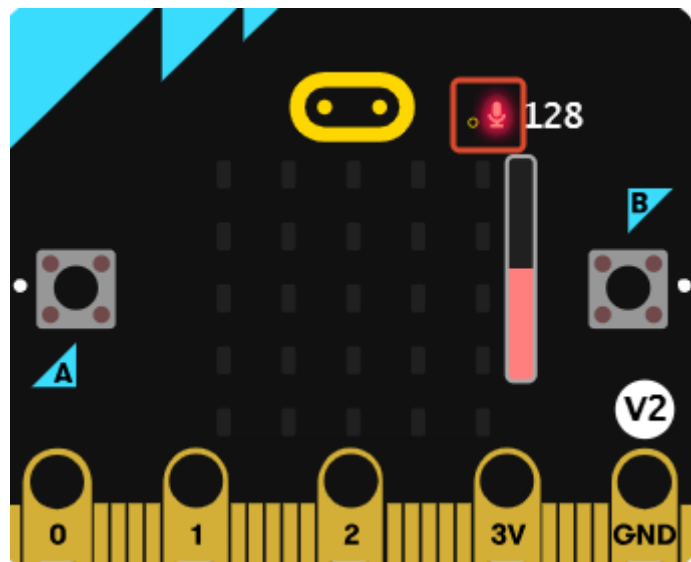
3. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. **Micro:bit V2 version** flashes `5.Scare_You_v2.hex`, **Micro:bit V1 version** flashes `5.Scare_You_v1v2.hex`. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

4. Experiment Phenomenon

The car will continuously detect the ambient sound. When the sound is loud enough, the car will show a crying face and shake backward, showing a "scared" effect. **Placing the sound source close to the microphone recording position makes it easier to detect the sound, and you can also adjust the sound detection threshold according to the actual situation.**

The recording position of the V2 version is as follows.



The recording position of the V1V2 version is as follows.

